



University of Ha'il  
College of Computer Science and Engineering  
Department of Computer Science

CSCE 232 - Software Engineering  
Project Final Report

## Tour & Event Planner System

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## Abstract

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In this project, we analyzed and designed the Tour and Event Planner System, which aims to provide a unified platform that helps users discover, organize, and book tourism events and activities across different regions of the Kingdom of Saudi Arabia.

Through conducting interviews and distributing questionnaires, several challenges faced by users were identified. These challenges included fragmented information, difficulties in booking, unclear event details, and limited coordination between event organizers and users. Based on the collected data, both functional and non-functional requirements were identified to address these issues and improve the overall user experience.

The project successfully reached the analysis and design stages, where system requirements were documented, and use case diagrams and sequence diagrams were developed. This work represents a strong foundation that can be used for future system implementation.

Despite some challenges such as limited time and difficulties in arranging interviews, the project objectives were successfully achieved. Overall, this project enhanced our understanding of the software development process and contributed to supporting the goals of Saudi Vision 2030 in the tourism and events sector.

## Acknowledgment

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We would like to express our sincere gratitude and appreciation to Dr. Lamia Al-Harbi for her continuous support and valuable guidance throughout the project implementation. She was always keen to clarify project requirements, correct mistakes, and provide constructive feedback that significantly helped us improve the quality of our work.

We also extend our sincere thanks to the employee of Amanat Hail Region, Mdar Travel and Tourism Agency, and all tourists and participants who took part in the interviews and questionnaire. Their time, information, and insightful feedback directly contributed to understanding user needs and defining the system requirements.

Finally, we would like to thank everyone who contributed to the success of this project, either directly or indirectly.

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## Project Plan and Task Distribution among Team Members

| <b>Week</b>        | <b>Task</b>                                                                              | <b>Team Member</b>           |
|--------------------|------------------------------------------------------------------------------------------|------------------------------|
| <b>Week 10, 11</b> | system description, goals, objectives, benefits, and environment Functional requirements | Fajr                         |
| <b>Week 10, 11</b> | Stakeholders of the System                                                               | Wasan                        |
| <b>Week 10, 11</b> | Interviews                                                                               | All Team Members             |
| <b>Week 10, 11</b> | Questionnaire                                                                            | Wasan - Haifa                |
| <b>Week 10, 11</b> | functional requirements                                                                  | Haifa - Wasan                |
| <b>Week 10, 11</b> | Nonfunctional requirements                                                               | Jana - Sarah                 |
| <b>Week 12</b>     | Four Fully dressed use case                                                              | Fajr - Sarah<br>Jana - Wasan |
| <b>Week 12</b>     | Use case Diagram                                                                         | Haifa                        |
| <b>Week 13</b>     | Sequence diagram                                                                         | Fajr                         |
| <b>Week 13</b>     | Class diagram (Bonus)                                                                    | All Team Members             |
| <b>Week 14</b>     | Initial Interface Design                                                                 | Sarah                        |
| <b>Week 14</b>     | Conclusion and Future Work                                                               | Wasan - Jana                 |
| <b>Week 14</b>     | Final Project Report Preparation                                                         | Wasan - haifa                |
| <b>Week 15</b>     | Revise document and prepare presentation                                                 | All Team Members             |

# Chapter 1: Introduction

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## 1.1. Project Description

As we have observed, it's clear that many event and tour companies here in Saudi Arabia like in many parts of the world — still rely on old, disconnected digital tools in their daily operations. This remains a major challenge, especially when considering how rapidly our tourism and entertainment sectors are growing under the Kingdom's Vision 2030.

What we constantly see is how travelers and even people living here in the Kingdom end up facing so many unnecessary roadblocks. They're forced to dig through tons of scattered info and jump between countless booking sites just to put together a simple trip or organize some activities. Honestly, this broken process isn't just a waste of valuable time, it's genuinely frustrating. It often discourages people from even trying to discover all the amazing, unique cultural, historical, and fun experiences KSA has to offer.

## 1.2. Goals and Objectives

first goal is to connect the user with a wide range of events and activities. Provide a smart search engine that allows the user to filter the search by location, price, time, and activity type. Provide complete, updated information about activities and display user reviews.



The second goal is to facilitate the booking and payment process. Support secure and multiple payment options such as credit cards, Apple Pay, and Google Pay to complete the payment process. Send booking confirmations and digital tickets via email and within the system interface.

The third goal is to organize and store travel itineraries/schedules. Creating and saving itineraries for multiple scenarios/cases and modifying them easily. Provide a notification feature when the trip date approaches or if a change occurs to it.

### **1.3. Purpose**

The project aims to facilitate travel planning by analyzing, designing, testing, deploying, maintaining, and updating an integrated tour and event planning system. This system will enable users to discover tourist activities and recreational events, make instant and reliable bookings, and organize and manage tour schedule

### **1.4. Environment**

The scope of this project defines the development of a system that serves three main types of users: the end user, the service provider(vendor) and, the administrator(admin), each of whom has their own account within it.

The admin- is responsible for the overall management of the system, overseeing service provider privileges, managing user accounts, and resolving related issues. the end user-Allows them to access available events, book them, manage payments, and organize personal travel itineraries the service provider(vendor) - Allows them to display their activities and tours, update their availability, review their upcoming bookings, and manage the basic data of the activity

## 1.5. Definitions

| Definition                       | Term                       |
|----------------------------------|----------------------------|
| Database                         | MySQL (v8.0)               |
| Database web management          | PhpMyAdmin (v4.2.9)        |
| Operating system                 | Ubuntu/Linux               |
| Web server                       | Nginx                      |
| Server-side programming language | PHP (v8.2)                 |
| Framework                        | Laravel (v10)              |
| Client-side programming language | JavaScript                 |
| Libraries                        | JQuery, Bootstrap (v3.2.0) |

## Chapter 2: Requirements Description

---

### 2.1. Stakeholders of the System

| Stakeholder / User      | Role                                                                      |
|-------------------------|---------------------------------------------------------------------------|
| Project Manager         | Leads the system work and ensures everything is finished on time.         |
| System Analyst          | Collects user needs and writes clear system requirements.                 |
| Developers / Designers  | Build and design the system, including event booking and tour features.   |
| Event Organizers        | Adds and updates events and ensures details and schedules are correct..   |
| Tour Company Managers   | Provides tour plans, prices, and availability.                            |
| Tourists / Users        | Use the system to search, book, and enjoy tours and events.               |
| Saudi Tourism Authority | Supports tourism activities and ensures alignment with Vision 2030 goals. |

*Table 1: Stakeholders of the system*

## 2.2. Requirements Gathering Methods

### 2.2.1 Interviews

#### Student Interviews: Fajr Al-bayih

##### Interview 1

**Interviewee Name:** Randa Al-Furaidi

**Date of Interview:** 3 November 2025

**Method:** In-person interview

| Question                                                                                                 | Answer                                                         |
|----------------------------------------------------------------------------------------------------------|----------------------------------------------------------------|
| What are the three most important criteria you consider when booking an event or activity?               | The price, the reviews, and that it's suitable for the family. |
| When searching for a specific activity, how do you prefer the results to appear — as a list or on a map? | I prefer them as a list.                                       |
| Would you like the application to include an?                                                            | Yes, that would be a great option.                             |

##### Interview 2

**Interviewee Name:** Roqayah Talal

**Date of Interview:** 3 November 2025

**Method:** In-person interview

| Question                                                                                   | Answer                                                                     |
|--------------------------------------------------------------------------------------------|----------------------------------------------------------------------------|
| What are the three most important criteria you consider when booking an event or activity? | That it's simple and clear, booking options and showing including reviews. |
| What payment methods do you prefer to have on the website?                                 | Apple Pay and Visa.                                                        |
| How long do you prefer to wait to receive the booking confirmation message?                | A maximum of one minute.                                                   |

### Interview 3

**Interviewee Name:** Gadah Al- gumadi

**Date of Interview:** 3 November 2025

**Method:** In-person interview

| Question                                                                                   | Answer                                   |
|--------------------------------------------------------------------------------------------|------------------------------------------|
| What are the three most important criteria you consider when booking an event or activity? | The location, Organization, and reviews. |
| Would you like to receive notifications when the event date is near?                       | yes.                                     |

### Student Interviews: Sarah Al-Amer

#### Interview 1

**Interviewee Name:** Participant 1

**Date of Interview:** 4 November 2025

**Method:** In-person interview

| Question                                                                                                  |                                      |
|-----------------------------------------------------------------------------------------------------------|--------------------------------------|
| When searching for an event, do you prefer the system to show only free activities or all available ones? | All available activities             |
| Do you prefer to pay directly through the app or at the event location?                                   | Direct payments through the website. |
| Would you like the system to allow easy searching of events with your friends?                            | Yes.                                 |
| Which language do you prefer for the app interface                                                        | Both English and Arabic.             |

#### Interview 2

**Interviewee Name:** Participant 2

**Date of Interview:** 4 November 2025

**Method:** In-person interview

| Question                                                                                                  | Answer                               |
|-----------------------------------------------------------------------------------------------------------|--------------------------------------|
| When searching for an event, do you prefer the system to show only free activities or all available ones? | Only f activities.                   |
| Do you prefer to pay directly through the app or at the event location?                                   | Direct payment through the website . |
| What matters most to you when choosing an event? (Price, rating, location, type of activity)?             | Price and ratings                    |
| Would you like the system to allow easy searching of events with your friends?                            | Yes .                                |
| Which language do you prefer for the app interface                                                        | Both English and Arabic.             |

### Interview 3

**Interviewee Name:** Participant 3

**Date of Interview:** 4 November 2025

**Method:** In-person interview

| Question                                                                                      | Answer                |
|-----------------------------------------------------------------------------------------------|-----------------------|
| Do you prefer to pay directly through the app or at the event location?                       | through the app .     |
| What matters most to you when choosing an event? (Price, rating, location, type of activity)? | Type of activity only |
| Would you like the system to allow easy searching of events with your friends?                | Yes .                 |
| Which language do you prefer for the app interface                                            | Arabic.               |

### Student Interviews: Haifa Al-fouzan

#### Interview 1

**Interviewee Name:** Mohammed Al-awlaki

**Nationality:** Emirati

**Purpose of visit:** Attending the Hail International Arabian horse Beauty Championship.

**Date of Interview:** 7 November 2025

**Method:** Face-to-face

| Question                                                                                                                 | Answer                                                            |
|--------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------|
| What motivated you to visit Ha'il and attend the Arabian Horse Beauty Championship?                                      | Passion for Arabian horses and exploring local culture.           |
| Did you face any difficulties while searching for information or booking the event or accommodation?                     | Yes, information was scattered and hard to find.                  |
| What main feature would you like to see in an application that gathers all tourism and event activities in Saudi Arabia? | Clear details about accommodation, user reviews, event reminders. |

## Interview 2

**Interviewee Name:** Ahmed AL - Amar

**Nationality:** Syrian

**Purpose of visit:** visiting Hail to explore its beautiful mountains and experience its tourism events.

**Date of Interview:** 7 November 2025

**Method:** Face-to-face

| Question                                                                | Answer                                                                                    |
|-------------------------------------------------------------------------|-------------------------------------------------------------------------------------------|
| What is the reason for your visit to Hail?                              | To explore Hail's beautiful mountains and rich heritage.                                  |
| Did you face any difficulties while planning your trip or upon arrival? | Yes, I didn't know how to identify heritage sites or events properly.                     |
| Are there any tourism programs that help you discover Hail?             | Existing programs lack details, pictures, and visitor recommendations.                    |
| What features did you need during your trip?                            | A website or app providing all details about heritage sites, events, hotels, restaurants. |

## Interview 3

**Interviewee Name:** Modhi Abdulaziz

**Method: Purpose of visit:** Event Supervisor & Media coordinator, Hail Municipality experience It's tourism events.

**Date of Interview:** 10 November 2025

**Method:** Face-to-face

| Question                                                             | Answer                                                                                                         |
|----------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------|
| What are the main challenges you face when organizing events?        | Coordinating participating entities and ensuring the audience receives clear information.                      |
| Do you face difficulties in communicating with visitors?             | Sometimes visitors don't know the exact locations or timings and need extra information.                       |
| What features would you like the event management system to include? | Notifications for attendees, clear display of locations and details, and fast communication with the audience. |

### **Student Interviews:** Jana Al-juhani

#### **Interview 1**

**Interviewee Name:** Mohammed Al-Johani

**Date of Interview:** 9 November 2025

**Method:** Face-to-face

| Question                                       | Answer                         |
|------------------------------------------------|--------------------------------|
| Preferred Payment Method                       | Direct payment via the website |
| Most Important Factors in Choosing an Activity | Rating and activity type.      |
| Sharing Activities Feature                     | yes                            |
| Preferred App Language                         | English and Arabic             |

#### **Interview 2**

**Interviewee Name:** Shahad Al-Harbi

**Date of Interview:** 9 November 2025

**Method:** Face-to-face

| Question                                       | Answer                         |
|------------------------------------------------|--------------------------------|
| Preferred Payment Method                       | Direct payment via the website |
| Most Important Factors in Choosing an Activity | Price and rating.              |
| Sharing Activities Feature                     | yes                            |
| Preferred App Language                         | English and Arabic             |



### Interview 3

**Interviewee Name:** Nagham Al-Harbi

**Date of Interview:** 9 November 2025

**Method:** Face-to-face

| Question                                       | Answer             |
|------------------------------------------------|--------------------|
| Preferred Payment Method                       | Through the app    |
| Most Important Factors in Choosing an Activity | Activity type only |
| Sharing Activities Feature                     | yes                |
| Preferred App Language                         | Arabic             |

### Student Interviews: Wasan Al-gharbi

#### Interview 1

**Interviewee Name:** Amanat Hail Employee

**Date of Interview:** 10 November 2025

**Method:** Face-to-face

| Question                                                                                                                   | Answer                                                                                                                                                                                            |
|----------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| What is the role of Amanat Hail in supporting and organizing events within the city?                                       | Amanat Hail contributes to preparing suitable locations for events, providing municipal services, and coordinating with organizers to ensure events are conducted in a safe and organized manner. |
| Do you think having a unified electronic system for organizing tourism events would improve coordination between entities? | Yes, a unified system would help organize information between entities, reduce delays, and make event details more accessible.                                                                    |
| What important feature should be included in a system that supports governmental entities in tourism and events?           | The system should be easy to use, provide accurate schedules and locations, and enhance coordination between governmental entities.                                                               |

## Interview 2

**Interviewee Name:** Mdar Travel and Tourism Agency

**Date of Interview:** 11 November 2025

**Method:** Online interview (WhatsApp)

| Question                                                                            | Answer                                                                                                                       |
|-------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------|
| What services does your agency currently provide?                                   | The agency provides travel-related services such as flight bookings, hotel reservations, and tourism arrangements.           |
| Would a unified system for tourism events and activities be useful for your agency? | Yes, it would be very useful, especially if it offers competitive prices and includes multiple destinations in one platform. |
| What features would you like to see in such a system?                               | Easy search functionality, competitive pricing, and direct communication with customers.                                     |

## Interview 3

**Interviewee Name:** Abeer Salem

**Date of Interview:** 7 November 2025

**Method:** Face-to-face

| The opQuestion                                                              | Answer                                                                                                                                               |
|-----------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------|
| What difficulties do you face when searching for or booking tourism events? | Difficulty in booking, unclear information about prices, locations, event duration, and whether the event quality matches the paid price.            |
| What features would you like to see in a tourism events application?        | Ease of use, flexible booking and cancellation, visitor organization to avoid overcrowding, and clear presentation of events using images or videos. |
| What makes you trust and use a tourism application?                         | Credibility, good organization, system reliability, and user ratings and reviews.                                                                    |

### 2.2.2 Questionnaires

#### I. Preparation

The questionnaire was prepared to collect information about users' needs, expectations, and the challenges they face when searching for tourism events and activities. The questions focused on identifying common difficulties experienced by users, the features they expect the system to provide, as well as their behavior during the booking process. The questions were designed in a clear and simple manner to ensure easy understanding and to encourage participants to provide accurate and reliable responses.

To view the questionnaire used in this study, please refer to the following link: Tourism Survey Form: (Insert the questionnaire link here)

#### II. Data Collection

The questionnaire was distributed online to participants from the Hail region. Participants were selected based on accessibility and relevance to the study in order to gather meaningful insights related to tourism activities. Although the questionnaire targeted residents of Hail, the collected data represents general user needs. Therefore, the proposed system is intended to support tourism activities across the Kingdom of Saudi Arabia.

### III. Results

The questionnaires and interviews conducted in this study helped identify several challenges faced by users when searching for tourism events and activities. One of the main challenges was the difficulty in accessing accurate and up-to-date information, such as event details, locations, prices, and available services, due to information being scattered across multiple platforms.

The results also showed a clear need for a unified system that allows users to easily explore events and filter results based on their preferences, such as price, location, and category. Many participants emphasized the importance of having a simple booking process supported by clear confirmation messages after completing a booking.

In addition, the results highlighted the importance of features such as event reminder notifications, interactive maps, and flexible payment options. Users preferred the availability of multiple payment methods, such as Apple Pay and Visa, due to the convenience and reliability they offer during the booking process.

Overall, the results confirmed that users are looking for a reliable and user-friendly system that provides an integrated platform to simplify the planning and booking of tourism events and activities.

#### IV. Analysis

Based on the results obtained from the questionnaires and interviews, it is evident that users face a fundamental issue related to fragmented information and the absence of a centralized platform that brings together tourism events and activities in one place. This fragmentation directly affects users' ability to plan efficiently and make informed decisions.

The analysis of participants' responses explains why high priority was given to functional requirements such as search, filtering, booking, and payment during the design of the proposed system. Features such as interactive maps, reminder notifications, and ratings address the identified challenges and significantly contribute to improving the overall user experience.

The strong focus on usability and accessibility reflects users' expectations for a simple and effective system. Although data collection targeted participants from the Hail region, the analysis indicates that these requirements are applicable on a broader scale. Therefore, the proposed system was designed to support tourism activities across different regions of the Kingdom of Saudi Arabia in alignment with Saudi Vision 2030.

## 2.3. Functional Requirements

| ID  | Requirement Description                                                                                                   | Priority | Acceptance Criteria                                                             |
|-----|---------------------------------------------------------------------------------------------------------------------------|----------|---------------------------------------------------------------------------------|
| R1  | The system shall allow users to create a new account using personal information such as name and email.                   | High     | The user account is successfully created and stored in the system database.     |
| R2  | The system shall allow users to log in and log out of their accounts securely.                                            | High     | The user can access the system after login and exit safely after logout.        |
| R3  | The system shall allow users to search for events and activities by name or location.                                     | High     | Relevant events are displayed based on the search criteria entered by the user. |
| R4  | The system shall allow users to filter events by price, rating, category, or date.                                        | High     | Filtered results are updated correctly according to the selected filters.       |
| R5  | The system shall allow users to view detailed information about events, including location, time, description, and price. | High     | Complete and accurate event details are displayed on the event details page.    |
| R6  | The system shall allow users to book events and activities online.                                                        | High     | A confirmation message is displayed after completing a booking.                 |
| R7  | The system shall allow users to view, cancel, or modify their bookings.                                                   | Medium   | The booking status is updated correctly after modification or cancellation.     |
| R8  | The system shall allow users to save events to a favorites list.                                                          | Medium   | Selected events appear in the user's favorites list.                            |
| R9  | The system shall send notifications to users before the event date.                                                       | Medium   | A notification is sent to the user at the scheduled time before the event.      |
| R10 | The system shall display event locations on an interactive map and show nearby landmarks.                                 | Medium   | Event locations are displayed accurately on the interactive map.                |
| R11 | The system shall allow users to view ratings and reviews from other users.                                                | Low      | Ratings and reviews are visible on each event page.                             |
| R12 | The system shall allow users to make payments using multiple methods such as Apple Pay and Visa.                          | High     | The payment is processed successfully and confirmed to the user.                |
| R13 | The system shall generate and send an electronic booking receipt after payment.                                           | Medium   | The user receives a digital receipt for the completed booking.                  |

Table 2: Functional Requirements

## 2.4. Non-Functional Requirements

| ID   | Non-Functional Requirement Description                                                                             | Category          | Priority | Acceptance Criteria                                                              |
|------|--------------------------------------------------------------------------------------------------------------------|-------------------|----------|----------------------------------------------------------------------------------|
| NFR1 | The system should provide fast response times and handle multiple users smoothly without freezing or slowing down. | Performance       | High     | Pages load within a few seconds and do not freeze under normal usage.            |
| NFR2 | The system must protect user data using encryption and allow access only to authorized users.                      | Security          | High     | Only authorized users can access system data, and unauthorized access is denied. |
| NFR3 | The system should be easy to use and navigate, with clear design suitable for all users.                           | Usability         | Medium   | Users can navigate and use the system without training or confusion.             |
| NFR4 | The system should operate reliably with minimal downtime and automatic data backup.                                | Reliability       | High     | The system operates without crashes and data is backed up automatically.         |
| NFR5 | The system should be available 24/7, with prior notification in case of maintenance.                               | Availability      | High     | The system is accessible 24/7 except during scheduled maintenance.               |
| NFR6 | The system should support future growth in users, events, and additional features.                                 | Scalability       | Medium   | The system supports increased users and activities without performance issues.   |
| NFR7 | The system should work on different browsers and devices and support multiple payment methods.                     | Compatibility     | Medium   | The system works correctly on different browsers, devices, and payment methods.  |
| NFR8 | The system must comply with Saudi data protection and privacy regulations.                                         | Legal and Ethical | High     | The system follows Saudi data protection and privacy regulations.                |
| NFR9 | The system should be easy to maintain, update, and fix without affecting functionality.                            | Maintenance       | Medium   | System updates and fixes can be done without affecting users or data.            |

Table 3: Non-Functional Requirements

## Chapter 3: Design Description

---

### 3.1. Use Cases

#### 3.1.1. UC01

|                |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |
|----------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Use Case ID    | UC 01                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |
| Use Case Name  | Login To System                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |
| Primary Actor  | End user , service provider, admin                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |
| Preconditions  | The actor must have a registered and activated account in the system                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |
| Postconditions | The actor is successfully logged in and directed to the appropriate interface for their role.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |
| Basic Flow     | <p>1. The Actor navigates to the Login Page.</p> <p>2.The System displays the login form with username/email and password fields.</p> <p>3.The Actor enters their username/email and password. 4. The Actor clicks the "Login" button.</p> <p>5.The System verifies the entered credentials.</p> <p>6. The System logs the actor in and directs them to their interface.</p>                                                                                                                                                                                                                                                                                                                                                                                                                  |
| Alternate Flow | <p>4.a.The actor entered an invalid username/email address.</p> <p>4.a.1. The system prompts the actor to enter a valid username/email address.</p> <p>4.a.2. he actor enters a valid username/email address. The use case resumes at Basic Flow Step</p> <p>4. 4.b. The actor entered an invalid password.</p> <p>4.b.1. The system prompts the actor to enter a valid password.</p> <p>4.b.2. The actor enters a valid password. The use case resumes at Basic Flow Step</p> <p>4.5.a. The actor entered an incorrect combination of username/email address and password.</p> <p>5.a.1.The system displays an error message stating that the combination is invalid.</p> <p>5.a.2. The actor makes another attempt to enter the credentials. The use case resumes at Basic Flow Step 4.</p> |
| Priority       | high                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |



### 3.1.2 UC02

|                |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |
|----------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Use Case ID    | UC 02                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |
| Use Case Name  | Search for events                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |
| Primary Actor  | Tourist (user)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |
| Preconditions  | UC 01 - Login To System , the user must be having a registered and activated account in the system                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |
| Postconditions | The system shows the events that match the user's search.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |
| Basic Flow     | <p>1.The system checks the request and displays matching events.</p> <p>2.The user types a search word or chooses filters like price or location.</p> <p>3.The user clicks the “search” button.</p> <p>4.The system checks the search request.</p> <p>5.The system shows the matching events.</p> <p>6.The use case ends.</p>                                                                                                                                                                                                                                                                                                                                                                                             |
| Alternate Flow | <p>A1 - (no results found)</p> <p>3.a the system finds no events that match the search or filters.</p> <p>3.a.1 It shows a short message (no events found).</p> <p>3.a.2 the user update the search and try again.</p> <p>A2 ( Invalid filters )</p> <p>3.b the user selects filter values that are not correct or do not match.</p> <p>3.b.1 The system asks the user to adjust the filters.</p> <p>3.b.2 The user fixes the filters and continues the search.</p> <p>A3 (search failed )</p> <p>5.a The system cannot load the results due to a slow connection or temporary issue.</p> <p>5.a.1 It shows ( Unable to load events right now, please try again ).</p> <p>5.a.2 The user may wait and try again later</p> |
| Priority       | high                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |

### 3.1.2. UC03

|                |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |
|----------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Use Case ID    | UC 03                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |
| Use Case Name  | Browse and Book Events                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |
| Primary Actor  | User (Traveler)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |
| Preconditions  | The user has a valid account and is logged into the system. The system has updated information about available tours and events.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |
| Postconditions | The user successfully views event details and completes a booking. The booking is saved in the user trip schedule.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |
| Basic Flow     | <ol style="list-style-type: none"><li>1. The user opens the Events &amp; Tours section.</li><li>2. The system displays a list of available events.</li><li>3. The user filters or searches for specific events.</li><li>4. The user selects an event to view more details.</li><li>5. The system shows detailed information such as date, location, price, and available seats.</li><li>6. The user clicks the Book Now option.</li><li>7. The system confirms availability and processes the booking.</li><li>8. The booking is added to the user itinerary.</li><li>9. The system shows a confirmation message.</li></ol> |
| Alternate Flow | <p>A1- 1. In Step 7 of Basic Flow, system finds no available seats.</p> <ol style="list-style-type: none"><li>2. System displays a message that the event is fully booked.</li><li>3. User chooses another event or returns to main list.</li></ol> <p>A2- 1. The user decides not to continue in Step 4 or 5.</p> <ol style="list-style-type: none"><li>2. The user returns to the event list without making a booking.</li></ol>                                                                                                                                                                                          |
| Priority       | high                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |

### 3.1.3. UC04

|                |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |
|----------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Use Case ID    | UC 04                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |
| Use Case Name  | View Event Information                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |
| Primary Actor  | Tourist (User)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |
| Preconditions  | UC 03 - Create User Account                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |
| Postconditions | The system displays the complete event details to the user.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |
| Basic Flow     | <ol style="list-style-type: none"><li>1. The user opens the Events page in the system.</li><li>2. The system displays a list of available events.</li><li>3. The user selects one event from the list.</li><li>4. The system retrieves the event details.</li><li>5. The system displays the event details (name, description, date, time, location, price, and organizer).</li><li>6. The use case ends.</li></ol>                                                                                                                                                                                                     |
| Alternate Flow | <p>A1 – Event Not Available:</p> <ol style="list-style-type: none"><li>4.a The selected event is no longer available.</li><li>4.a.1 The system informs the user that the event is no longer available.</li><li>4.a.2 The system returns the user to the list of available events.</li></ol> <p>A2 – Connection Error:</p> <ol style="list-style-type: none"><li>4.b The system is unable to load the event details due to a network issue.</li><li>4.b.1 The system displays: “Unable to load event details. Please try again.”</li><li>4.b.2 The use case ends or the user may retry from Basic Flow step 3.</li></ol> |
| Priority       | high                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |

### 3.2. Use Case Diagram

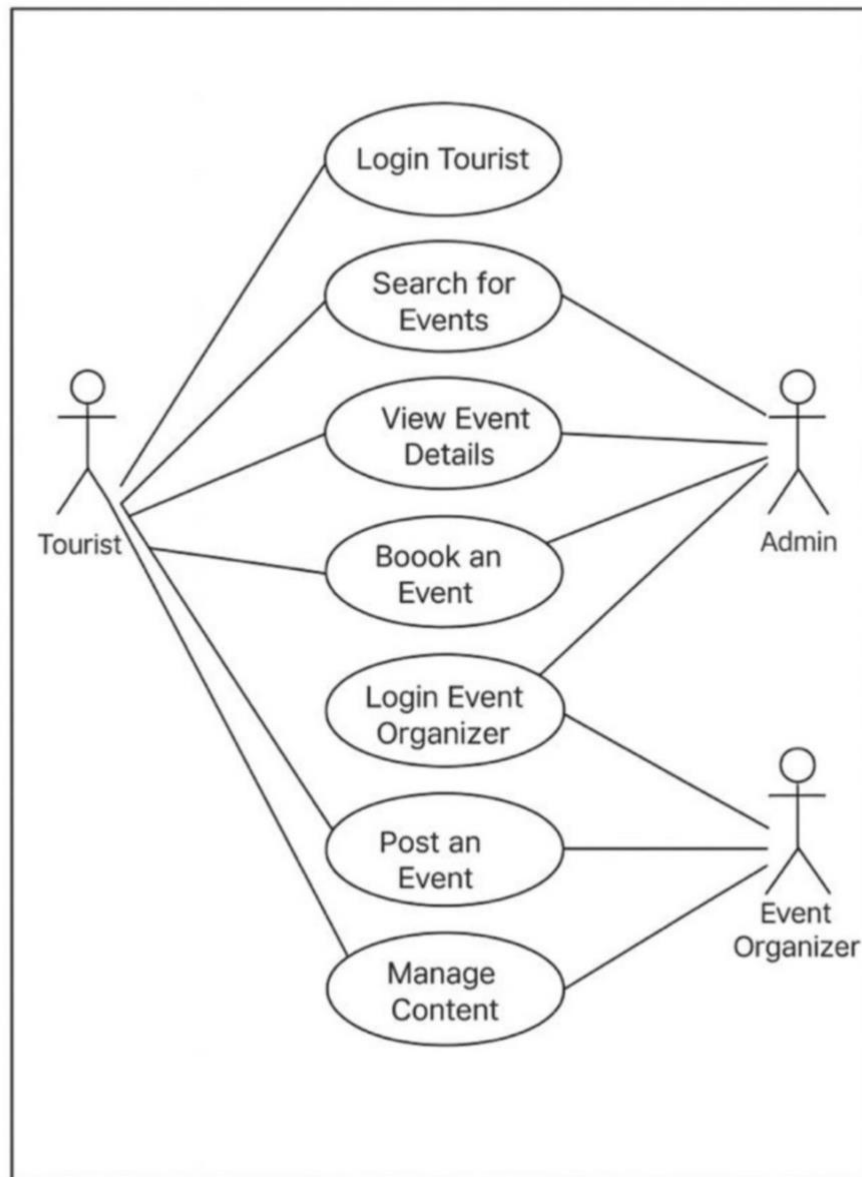


Figure 1: Use Case Diagram

### 3.3. System Sequence Diagram

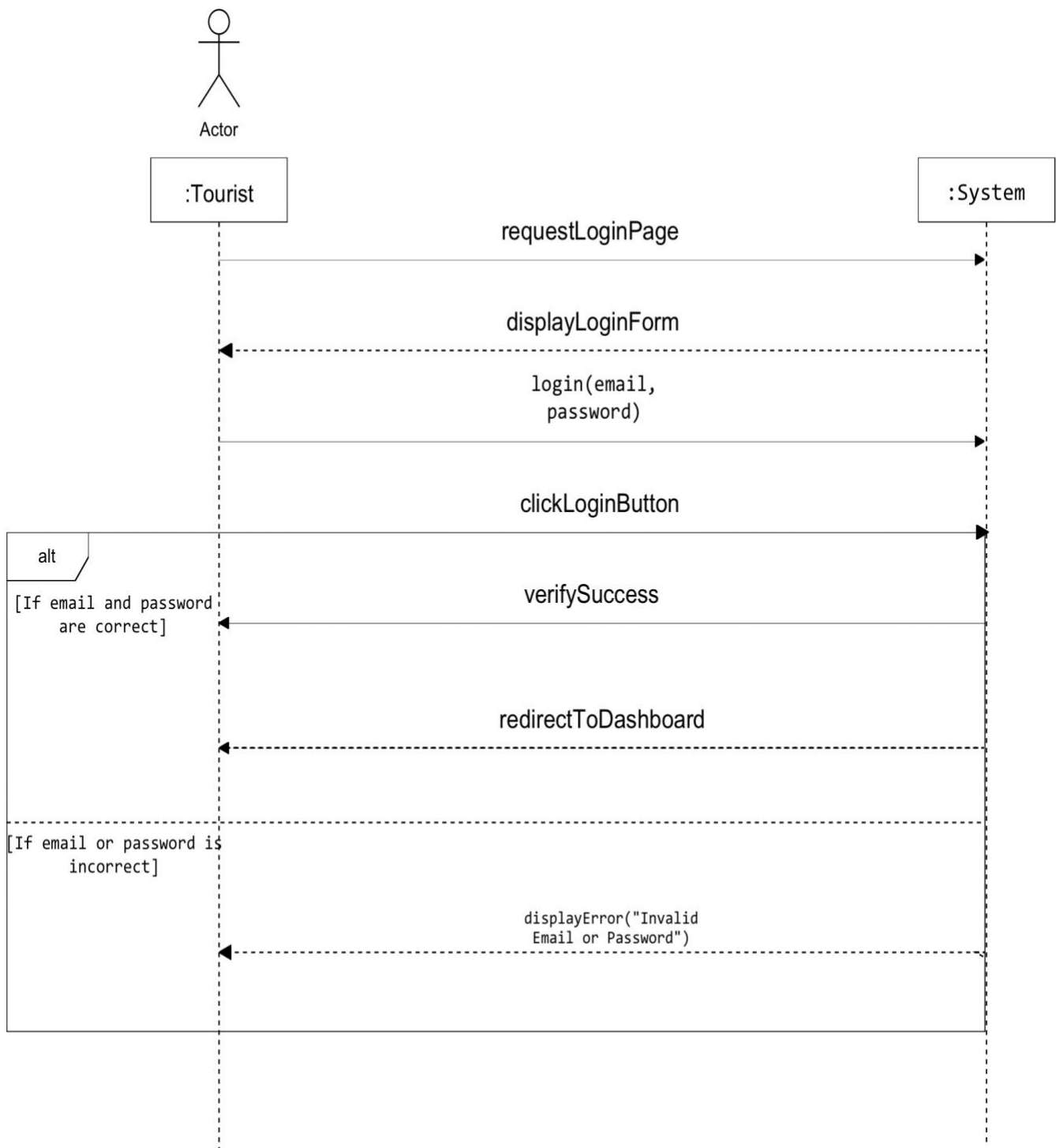


Figure 2: System Sequence Diagram

## **Participating Objects**

1- Tourist: (Actor)

2- System: (System)

## **Scenario**

The Tourist requests the login page.

The System displays the login form.

The Tourist enters email and password.

The Tourist clicks the login button.

### **Alternative (Alt) Frame:**

[If email and password are correct]:

The System verifies the credentials successfully and redirects the Tourist to the dashboard.

[If email or password is incorrect]:

The System displays an error message ("Invalid Email or Password").

### 3.4. Class Diagram – Tour & Event Planner System

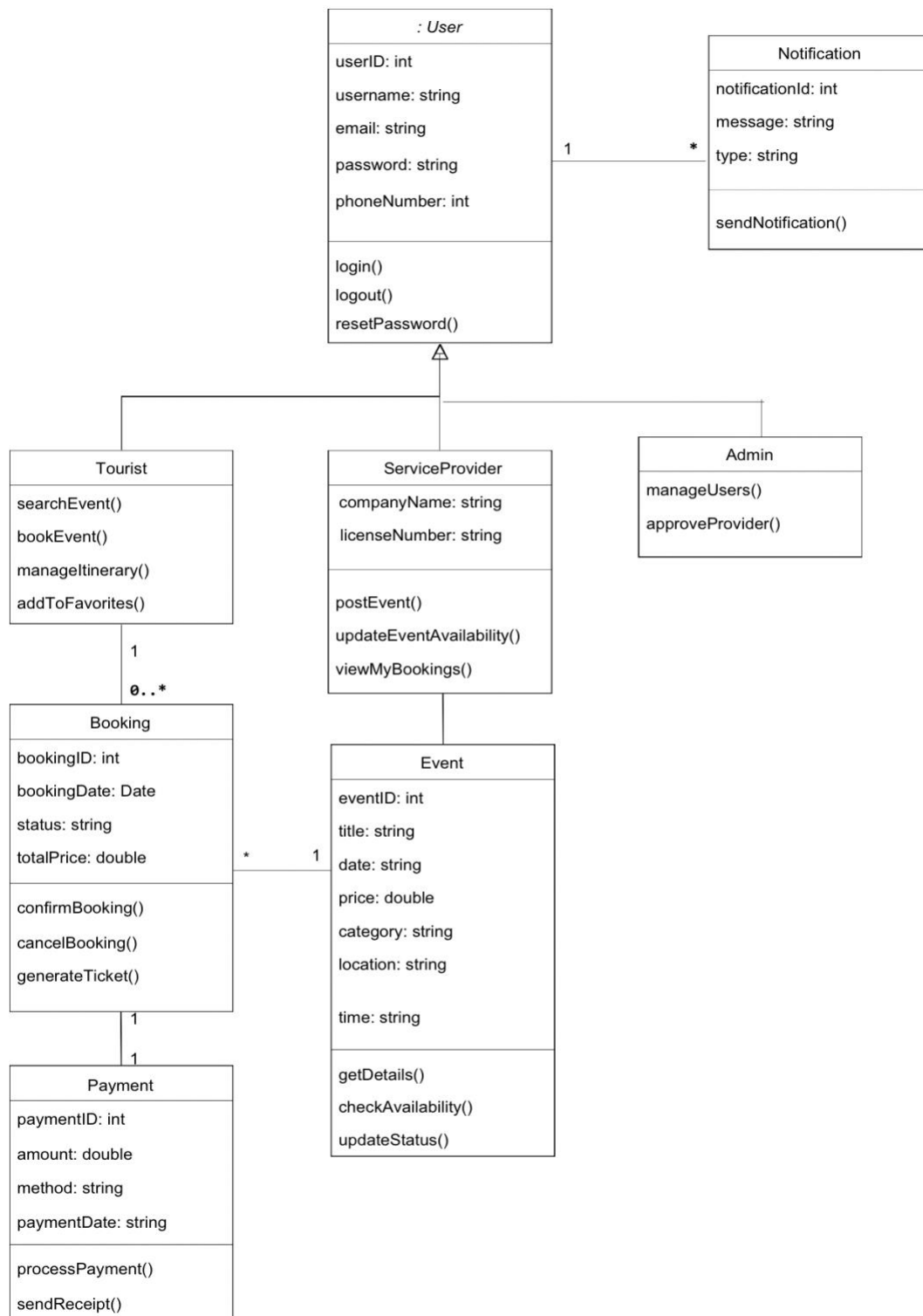


Figure 3: Class Diagram

## 3.5. Initial Interface Design

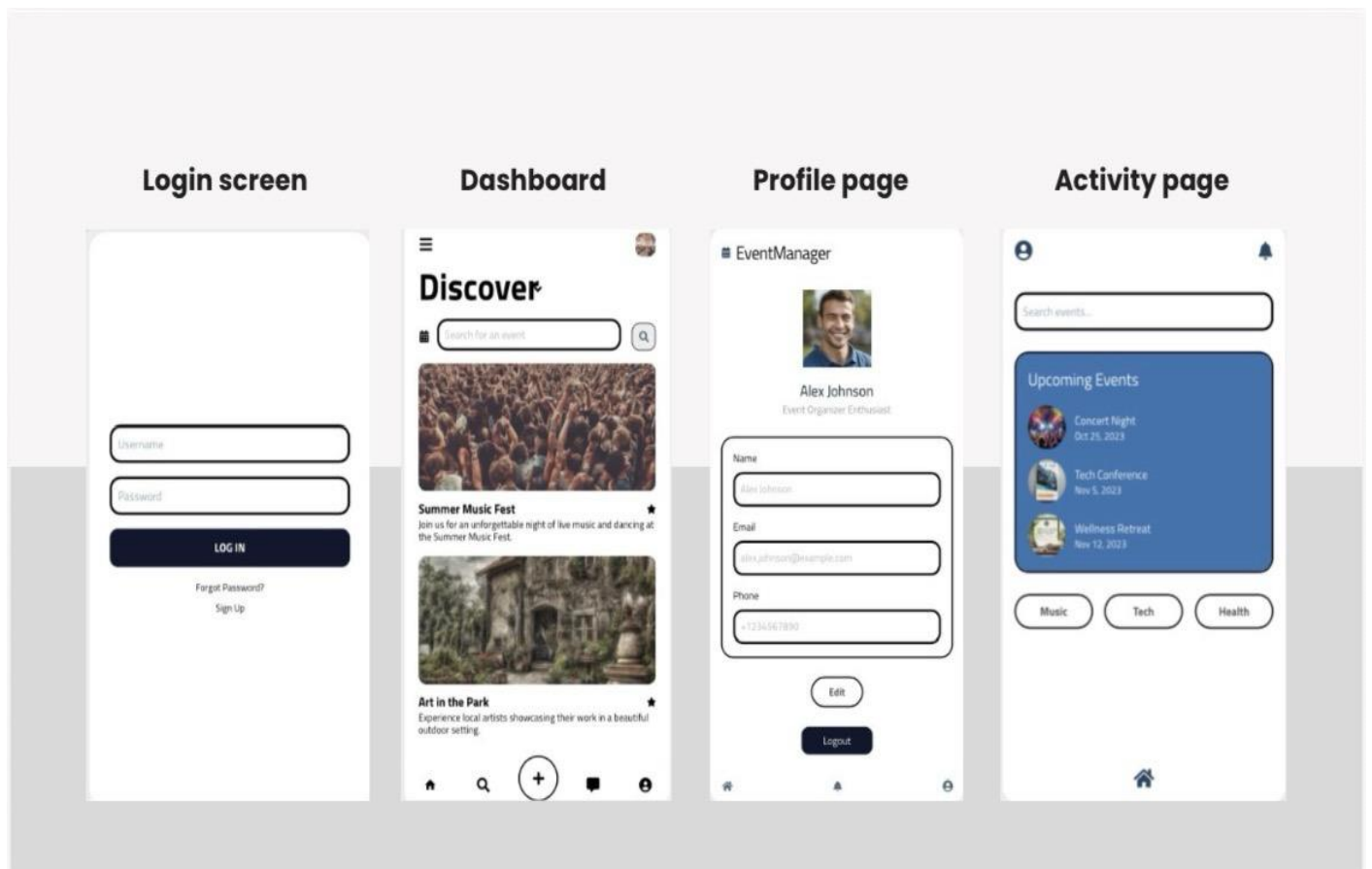


Figure 4: Initial Interface Design



## Main System Screens Description Table

*Table 4: Main System Screens Description*

| Screen Name   | Purpose of Screen                                     | Main Components (Buttons, Forms, Menus)                            | Available Actions                                      | Connection to System Requirements                      |
|---------------|-------------------------------------------------------|--------------------------------------------------------------------|--------------------------------------------------------|--------------------------------------------------------|
| Login Screen  | Allows users to log in to the system                  | Username field, Password field, Login button, Forgot Password link | Enter credentials, Click login                         | UC01 – Login to System                                 |
| Dashboard     | Provides overview of user activities and events       | Navigation menu, Summary panels, Notifications panel               | Navigate to events, view notifications, access profile | UC02 - Search for Events,<br><br>UC03 - Book an Event  |
| Profile Page  | Displays user personal information and settings       | Profile form, Edit button, Save button, Logout button              | View profile, Edit profile, Save changes               | UC01 – Login to System                                 |
| Activity Page | Shows booked events, upcoming activities, and history | Event list, Filter options, Calendar view, Details button          | View event details, Filter activities                  | UC03 - Book an Event,<br><br>UC04 - View Event Details |

## Chapter 4: Conclusion and Future Works

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### 4.1. Conclusion

In this project, we analyzed and designed the Tour & Event Planner System, which aims to provide a unified platform that helps users discover tourism events and activities, view details, and book trips easily.

By collecting user requirements through surveys and interviews, and by creating use cases and system diagrams, we were able to clearly define the main functions of the system. The proposed system helps reduce scattered information and improve the overall user experience.

Most of the project objectives were achieved, as the project reached a clear analysis and design stage that explains how the system should work. This provides a strong foundation for future implementation.

Some challenges were faced during the project, including limited time, difficulty in finding tourists for interviews, and coordinating with specialized entities. Despite these challenges, the project requirements were completed successfully.

Overall, the project achieved its goals and helped us better understand the software development process, while supporting tourism activities in line with Saudi Vision 2030.

## 4.2. Future Work

**The Tour & Event Planner System can be improved in future versions by adding features that enhance user experience and system efficiency, such as:**

1. Interactive Maps Integration: Adding maps to show event locations and tourist activities, along with suggested routes for easier navigation.
2. Notifications and Alerts: Sending reminders about upcoming events, schedule changes, or special offers.
3. Secure Online Payment Support: Enabling users to pay for events and activities safely within the application.
4. Ratings and Reviews System: Allowing users to rate events and leave feedback to help others make informed decisions.
5. Automated Trip Planner: Creating a complete travel schedule based on the user's available time and selected activities.
6. Offline Mode: Allowing users to view their itinerary and saved events even without an internet connection.
7. Multi-language Support: Adding multiple languages to make the system accessible to international tourists.

## Chapter 5: References:

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- [1] CSCE232 Course – Software Engineering, University of Hail.
- [2] CSCE232 Course Project Example – “TotStory” Daycare Center System, University of Hail.